

Task 10: Firewall Configuration & Testing

1. Introduction

A firewall is a network security mechanism used to monitor and control incoming and outgoing network traffic based on predefined security rules. Firewalls act as a barrier between trusted internal networks and untrusted external networks, such as the Internet. In this task, firewall concepts were studied, and basic firewall rules were configured and tested using system-level firewall tools.

2. Objectives

The main objectives of this task are:

- To understand firewall concepts and working principles
 - To configure basic firewall rules
 - To allow and deny specific ports
 - To test network connectivity after applying rules
 - To observe firewall logs
 - To block a suspicious or malicious IP address
 - To document firewall rules and analyze their impact
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3. Tools Used

- **UFW (Uncomplicated Firewall) / Windows Defender Firewall**
 - **iptables** (alternative tool)
 - Command line / terminal
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4. Firewall Concepts

A firewall filters traffic based on rules such as IP address, port number, protocol, and direction of traffic. Firewalls can be classified as:

- **Stateful firewall:** Tracks active connections and allows packets based on connection state
- **Stateless firewall:** Filters packets without tracking connection state

Firewalls help prevent unauthorized access, reduce attack surfaces, and protect systems from network-based attacks.

5. Firewall Rule Configuration

Firewall rules were configured to control network traffic. The following actions were performed:

- Default deny policy was enabled for incoming traffic
- Specific ports such as SSH (22) or HTTP (80) were allowed
- Unused or risky ports were blocked
- Rules were applied separately for inbound and outbound traffic

Each rule was verified after configuration to ensure correct behavior.

6. Allowing and Denying Ports

- Essential service ports were explicitly allowed
- All other unnecessary ports were denied
- This helped reduce exposure to unauthorized services

Example actions:

- Allow SSH for remote access

- Block unused ports to minimize attack surface
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7. Connectivity Testing

After configuring firewall rules:

- Network connectivity was tested using tools such as **ping** and service access checks
- Allowed services were accessible
- Blocked services were correctly denied

This confirmed that firewall rules were working as expected.

8. Firewall Logs Observation

Firewall logs were checked to:

- Monitor allowed and blocked traffic
- Identify denied connection attempts
- Understand how firewall decisions were made

Logs help in detecting suspicious activities and troubleshooting network issues.

9. Blocking Malicious IP Address

A test IP address was manually blocked using firewall rules.

After blocking:

- Connection attempts from that IP were denied
- Logs confirmed the blocking action

This demonstrates how firewalls can be used to mitigate attacks from known malicious sources.

10. Impact Analysis

Implementing firewall rules improved system security by:

- Restricting unauthorized access
- Allowing only required services
- Providing visibility into network traffic

However, improper firewall rules can also block legitimate services, so careful configuration is required.

11. Conclusion

This task provided hands-on experience with firewall configuration and testing. It helped in understanding how firewall rules protect systems, control traffic flow, and enhance overall network security. Firewall management is a critical skill for cybersecurity professionals.

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